

Two-Sided Spectral Control of Wilson One-Plaquette Class Gaps and an Exactly Verified Strong-Coupling Bridge to the SU(3) Glueball Channels

[Author]*

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Abstract

We study the one-plaquette class Hamiltonian $H_\beta = \frac{1}{2}C_2 + \beta(1 - \frac{1}{N}\text{ReTr}g)$ of SU(N) Wilson lattice gauge theory, restricted to gauge-invariant (class-function) sectors, and establish exact analytic control of its spectral gap from both ends of the coupling range. For SU(3) we prove a six-coefficient weak-coupling law (through order β^{-2} , all coefficients exact rationals or rational multiples of $\sqrt{6}$) and exact strong-coupling towers through β^{10} (charge-even) and β^6 (charge-odd); a branch-factored two-point Padé representation built from proven inputs alone is globally valid to 9.5×10^{-4} relative accuracy on $\beta \in [0.25, 50]$. For general N we give closed-form coefficients, a parity-rationality theorem, and an identification of the underlying Weyl–Gaussian measure with the traceless-GUE eigenvalue ensemble, which renders every matrix element of the program an exact Wick computation and yields a parameter-free, falsifiable observable $R_0(N) = 3(N^2 - 4)/(N(N^2 + 1)(N^2 + 3))$, confirmed by Monte Carlo at the half-percent level for SU(3). We then prove a *Bridge Theorem*: in the Kogut–Susskind Hamiltonian the plaquette-restricted glueball masses are exactly $m_\pm(y) = 4\Delta_{\text{SU}(3)}^\pm(3y/2)$, so the entire local stack transfers verbatim to the restricted 4D strong-coupling scalar channels. The first correction beyond restriction — the $O(y^2)$ neighbor leakage — is computed exactly in both charge sectors (per-neighbor self-energy $-481/612$, C -independent; C -even hopping $-481/612$; C -odd hopping $5s/612$ with shared-link orientation s), and every coefficient is verified by an independent exact diagonalization on a two-plaquette lattice, with all four predicted level coefficients (1769/3060, 13/20, 31/68, 17/36) reproduced to one part in 10^6 . A small Monte Carlo measurement at $\beta = 5.7$, $am(0^{++}) = 1.15(25)$, is consistent with the known finite- β value 0.987(9). This paper makes no claim about the continuum limit or the Yang–Mills mass gap; its scope is exact strong-coupling structure with a fully auditable verification chain.

1 Introduction

The strong-coupling regime of lattice gauge theory is one of the few places in non-abelian gauge dynamics where exact statements are possible, and the Hamiltonian formulation of Kogut and Susskind [1] reduces its leading structure to a single compact object: the one-plaquette class Hamiltonian

$$H_\beta = \frac{1}{2}C_2 + \beta\left(1 - \frac{1}{N}\text{ReTr}g\right), \quad g \in \text{SU}(N), \quad (1)$$

*Draft of June 11, 2026. Single-manuscript consolidation of the local spectral program; verification certificates listed in Appendix A.

acting on class functions (characters), with C_2 the quadratic Casimir (Laplace–Beltrami) operator. Its spectral gap $\Delta(\beta)$, in each charge-conjugation sector, is the elementary excitation energy of a single unit of chromoelectric flux, and it calibrates the plaquette degree of freedom that every Hamiltonian strong-coupling expansion, every tensor-network truncation, and every quantum-simulation encoding of lattice gauge theory is built upon.

This paper consolidates, in a single manuscript, a program of exact results about $\Delta^\pm(\beta)$ and about its precise relation to the 3+1-dimensional strong-coupling glueball channels. The results are of three kinds.

(I) Two-sided exact control of the local gap. For $SU(3)$ we prove the weak-coupling law through six orders (Theorem 2.1), the strong-coupling towers through β^{10} (charge-even) and β^6 (charge-odd) (Theorem 4.1), and a globally valid closed-form representation accurate to 9.5×10^{-4} across the entire crossover (Theorem 4.2). For general N we prove closed-form coefficients, a parity–rationality theorem (Theorem 2.2), and the structural identification of the underlying measure with the traceless-GUE eigenvalue ensemble (Proposition 3.1), from which a parameter-free falsifiable observable follows (Theorem 3.2).

(II) An exact bridge to the 4D glueball channels. Theorem 5.1 shows that the magnetic term of the Kogut–Susskind Hamiltonian, restricted to a single plaquette, reproduces the compact Hamiltonian (1) with a quadrupled Casimir weight, so that the restricted 4D glueball masses are *exactly* $m_\pm(y) = 4 \Delta^\pm(3y/2)$. Every coefficient of (I) is thereby a statement about the restricted 0^{++} and C -odd scalar channels. The first correction beyond restriction, the $O(y^2)$ neighbor leakage, is computed exactly in both charge sectors (Theorems 6.2 and 6.3), with structural by-products: the fused-channel norms are dimension-weighted ($N_R = d_R/9$, Lemma 6.1); the neighbor self-energy is independent of charge sector; the C -even hopping equals the self-energy by an operator identity; and the C -odd hopping is a small, orientation-signed amplitude, so the C -odd excitation is nearly immobile at strong coupling.

(III) Independent verification. Every claimed coefficient is confirmed by at least two fully independent methods. The $O(y^2)$ bridge constants are verified by an exact diagonalization of the Kogut–Susskind Hamiltonian on a two-plaquette lattice (the “domino”), carried out in exact rational arithmetic with no perturbative input; the four predicted level coefficients are reproduced to one part in 10^6 (Section 7). The verification chain — scripts, gates, and what each certifies — is inventoried in Appendix A.

1.1 Scope, and what this paper is not

Everything in this paper lives at strong coupling. The existence of a mass gap in that regime is classical [3]; our contribution is exact *structure*: closed-form coefficients, two-sided global control, structural identifications, and a verified dictionary between the local object and the 4D channels. Nothing here addresses the continuum limit. In particular, this paper makes *no* claim about the Yang–Mills mass gap: the strong-coupling expansions below have finite radii of convergence set by explicit level-repulsion singularities, the lattice corrections at accessible couplings are large (the well-known Wilson-action scalar dip is visible in our own Monte Carlo data, Section 8), and no resummation performed here reaches the scaling window. Open problems are listed as open in Section 9; none is represented as imminent.

1.2 Conventions

$g \in \text{SU}(N)$; $\chi \equiv \chi_F(g) = \text{Tr} g$ and $\bar{\chi} = \bar{\chi}$; irreducible characters $\chi_{(p,q)}$ for $\text{SU}(3)$. Charge conjugation acts by $g \mapsto \bar{g}$, i.e. $\chi \leftrightarrow \bar{\chi}$; the gap in the $C = \pm$ sector is $\Delta^\pm(\beta)$. Power sums on Weyl-chamber variables are $p_k = \sum_j x_j^k$. In Sections 5–7 the 3+1-dimensional Kogut–Susskind Hamiltonian is taken in the convention

$$h = \sum_{\text{links } \ell} \frac{1}{2} C_2(\ell) - y \sum_{\text{plaquettes } p} (\chi_p + \bar{\chi}_p), \quad (2)$$

with χ_p the fundamental character of the oriented plaquette holonomy.

2 Weak coupling: the six-coefficient law and general N

In the weak-coupling (large- β) regime the class Hamiltonian is analyzed in Weyl-chamber coordinates on the Cartan plane: after the Gaussian rescaling $\lambda^4 = N/(2\beta)$ the potential and the Weyl-measure factor organize into a graded perturbation series around an exactly solvable shell problem, whose matrix elements close under Wick calculus (Section 3). A graded Rayleigh–Schrödinger recursion in a rational second-quantized (free-fermion Fock) representation then produces exact coefficients; full derivations are given in the companion derivation files inventoried in Appendix A, and every coefficient below is additionally confirmed by an independent compact character-basis numerical extraction.

Theorem 2.1 ($\text{SU}(3)$ weak-coupling law, six exact coefficients). *In the charge-even sector,*

$$\begin{aligned} \Delta_{\text{SU}(3)}^+(\beta) = & \sqrt{\frac{2\beta}{3}} - \frac{5}{16} - \frac{311\sqrt{6}}{9216} \beta^{-1/2} - \frac{5665}{110592} \beta^{-1} \\ & - \frac{8470769\sqrt{6}}{509607936} \beta^{-3/2} - \frac{56673445}{1528823808} \beta^{-2} + O(\beta^{-5/2}). \end{aligned} \quad (3)$$

In the charge-odd sector the leading behavior is $\Delta^- = \sqrt{3\beta/2} + O(1)$, and the coefficients at orders β^{-1} and β^{-2} are

$$c_2^- = -\frac{53}{864}, \quad c_4^- = -\frac{290599777}{6115295232}. \quad (4)$$

Theorem 2.2 (General N : closed forms and parity–rationality). *Write $\Delta_{\text{SU}(N)}^+(\beta) = \sqrt{2\beta/N} + \sum_{j \geq 0} c_j^{(N)} \beta^{-j/2}$ (charge-even) and analogously for the charge-odd sector, whose leading term is $3\sqrt{\beta/(2N)}$. Then $c_j^{(N)} \in \mathbb{Q}$ for even j and $c_j^{(N)} \in \sqrt{2N} \mathbb{Q}$ for odd j . Closed forms include*

$$c_0^{(N),+} = -\frac{2N^2 - 3}{16N}, \quad (5)$$

$$c_2^{(N),+} = -\frac{60N^6 - 401N^4 + 1522N^2 - 2297}{49152 N^2}, \quad c_2^{(N),-} = -\frac{95N^6 - 981N^4 + 5853N^2 - 15335}{49152 N^2}, \quad (6)$$

$$c_3^{(N),+} = -\sqrt{2N} \frac{2970N^8 - 27878N^6 + 166512N^4 - 546024N^2 + 734405}{2^{21} \cdot 3^2 \cdot N^3}. \quad (7)$$

The $\text{SU}(3)$ specializations reproduce (3) exactly; the formulas are verified for $3 \leq N \leq 10$ by the independent rational Fock engine.

Remark 2.3 (The rank-two angular term). The coefficient $c_1^{(3),+} = -311\sqrt{6}/9216$ contains an irreducibly *angular* contribution $\sqrt{6}/576$ at order $\beta^{-1/2}$ that any radial (single-variable Laguerre-type) reduction of the Weyl measure misses. It is the cleanest local signature of the full rank-two geometry, and it is confirmed numerically at roughly $8000\times$ the fit residual of the character-basis extraction.

3 The measure is the traceless GUE

Proposition 3.1 (GUE identification). *After the Gaussian rescaling, the radial weight induced by the $SU(N)$ Weyl measure on the trace-zero Cartan plane is the eigenvalue measure of the traceless Gaussian Unitary Ensemble; in particular the radial density is $u^{\alpha_N} e^{-u}$ with*

$$\alpha_N = \frac{\dim \mathfrak{su}(N) - 2}{2} = \frac{N^2 - 3}{2}, \quad (8)$$

the first graded perturbation is universal,

$$H_1 = -\frac{p_4}{48} \quad \text{for every } N, \quad (9)$$

and the conditional projection constant $\mathbb{E}[p_4 | p_2 = u] = \gamma_N u^2$ has the closed form $\gamma_N = (2N^2 - 3)/(N(N^2 + 1))$.

The identification is the computational engine of the entire weak-coupling program: every matrix element required by the Rayleigh–Schrödinger recursion becomes an exact Wick contraction in the GUE ensemble, which is why the coefficients of Theorems 2.1–2.2 can be carried in exact arithmetic to high order.

Theorem 3.2 (Parameter-free moment-ratio observable). *Let $R = \langle p_3^2 \rangle / \langle p_2^3 \rangle$ in the one-plaquette Gibbs ensemble. Then*

$$R_0(N) = \frac{3(N^2 - 4)}{N(N^2 + 1)(N^2 + 3)}, \quad \frac{r_1}{R_0} = \frac{N^2(N^2 - 9)}{8(N^2 + 1)}, \quad (10)$$

where r_1 is the coefficient of β^{-1} . The first-order slope vanishes identically at $N = 3$, where the expansion begins at second order:

$$R_{SU(3)}(\beta) = \frac{1}{24} \left(1 - \frac{7}{80} \beta^{-2} + O(\beta^{-3}) \right). \quad (11)$$

Monte Carlo measurements at $\beta = 5.8\text{--}160$ confirm $R_{SU(3)} = 1/24$ at the half-percent level. The sharpest untested prediction of the family is the $SU(4)$ slope $r_1/R_0 = 14/17$; we record it here as a falsifiable target for any group, including ones with no stake in this program.

4 Strong coupling: exact towers and global closure

Theorem 4.1 (Strong-coupling towers). *Rayleigh–Schrödinger expansion of (1) about $\beta = 0$ in the irreducible-character basis gives, for $SU(3)$,*

$$\Delta^+(\beta) = \frac{2}{3} - \frac{\beta}{6} + \frac{13}{180}\beta^2 + \frac{101}{2700}\beta^3 + \frac{1657}{567000}\beta^4 - \frac{32657}{7620480}\beta^5 - \frac{2167017157}{800150400000}\beta^6 + \dots, \quad (12)$$

$$\Delta^-(\beta) = \frac{2}{3} + \frac{\beta}{6} + \frac{1}{18}\beta^2 + \frac{7}{432}\beta^3 + \frac{143}{181440}\beta^4 - \frac{34877}{15240960}\beta^5 - \frac{2055143}{1600300800}\beta^6 + \dots, \quad (13)$$

with the charge-even tower extended through β^{10} (the four further exact coefficients are listed in Appendix B).

The exact compact gap is non-monotonic, with a minimum near $\beta \approx 0.8$ produced by level repulsion against a descending even-fundamental state; this nearby complex singularity sets the small convergence radius of (12). Quantitatively, the strong tower is accurate to better than 2% for $\beta \leq 1.5$ and the six-term weak law to better than 0.7% for $\beta \geq 2.5$, leaving the seam $\beta \in (1.5, 2.5)$.

Two negative results delimit what closes that seam. A two-point Padé in $x = \sqrt{\beta}$, matched to both towers with a pole-free constraint, cannot do better than 6.1×10^{-2} relative error; and cubic algebraic approximants matched to both towers develop a genuine branch defect (the two positive real branches collide near $\beta \approx 4.7$). The level-repulsion structure defeats rational and low-order algebraic global forms.

Theorem 4.2 (Branch-factored global representation). *Writing $\Delta^+(\beta) = \sqrt{c + \frac{2}{3}\beta} g(\beta)$ with auxiliary scale $c = \frac{1}{2}$, and representing g by a pole-free two-point Padé [13/13] in $x = \sqrt{\beta}$ matched to the proven strong tower through β^{10} and the proven six weak orders, yields a closed-form Δ^+ with*

$$\max_{\beta \in [0.25, 50]} \left| \frac{\Delta_{\text{rep}}}{\Delta_{\text{exact}}} - 1 \right| = 9.5 \times 10^{-4}, \quad (14)$$

pole-free on $\beta \geq 0$ and exact in both limits by construction.

To our knowledge this is the first globally valid closed-form representation of a non-abelian one-plaquette class gap built entirely from proven series inputs.

5 The Bridge Theorem

We now connect the local object to the 3+1-dimensional theory (2). The unperturbed ($y = 0$) ground state is the chromoelectric vacuum; the lowest excitations are single-plaquette flux loops $\chi_p |0\rangle, \bar{\chi}_p |0\rangle$ at energy $2C_2(F) = 8/3$ (four links at $\frac{1}{2}C_2$ each).

Theorem 5.1 (Bridge Theorem, plaquette-restricted level). *Restrict the magnetic term of (2) to a single plaquette p . The flux tower on p is then exactly the compact one-plaquette class Hamiltonian with quadrupled Casimir weight,*

$$h_{\text{loop}} = 4[H_\beta - \beta] \Big|_{\beta=3y/2}, \quad (15)$$

and consequently the restricted glueball masses in the two charge sectors are

$$m_\pm(y) = 4 \Delta_{\text{SU}(3)}^\pm \left(\frac{3y}{2} \right). \quad (16)$$

Proof. The four links of the plaquette each carry $\frac{1}{2}C_2$ of the common loop representation, quadrupling the Casimir weight of (1); matching the magnetic normalizations, $-y(\chi + \bar{\chi}) = 4 \cdot (-\beta/6)(\chi + \bar{\chi})$ at $\beta = 3y/2$, and the additive constant cancels in gaps. $O(y)$ check: degenerate perturbation theory in $\{\chi|0\rangle, \bar{\chi}|0\rangle\}$, using $\int \chi^3 = 1$ and $\int \chi^2 \bar{\chi} = 0$, gives the matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, hence $m_+ = \frac{8}{3} - y$ and $m_- = \frac{8}{3} + y$, in exact agreement with (16) expanded to first order. $\square$

Every coefficient of Sections 2–4, and the global representation of Theorem 4.2, is therefore — verbatim, with the substitution $\beta = 3y/2$ and an overall factor 4 — a statement about the restricted 4D strong-coupling scalar channels.

6 Leakage at $O(y^2)$: exact coefficients in both sectors

Beyond restriction, the first correction comes from plaquettes p' sharing a link ℓ with p . After integrating the six free links by the fusion identity $\int dB \text{Tr}(XB) \overline{\text{Tr}(YB)} = \text{Tr}(XY^\dagger)/3$, every required second-order quantity reduces to balanced $(2, 2)$ Haar moments on the shared link, resolved into the irreducible channels of the fused link representation.

Lemma 6.1 (Dimension-weighted channel norms). *Let ψ be the product of the two fundamental plaquette characters sharing ℓ . Inserting V on the shared link gives $\langle \psi, V \cdot \psi \rangle = |\chi_F(V)|^2/9$ (opposite relative orientation) or $\chi_F(V)^2/9$ (same orientation), whence by Schur orthogonality the channel norms are*

$$N_R = \frac{d_R}{9} : \quad \{1 : \frac{1}{9}, 8 : \frac{8}{9}\} \quad \text{and} \quad \{\bar{3} : \frac{1}{3}, 6 : \frac{2}{3}\}, \quad (17)$$

with channel energies (above $E_0 = \frac{8}{3}$) given by the link Casimirs: $1 \mapsto 4$, $8 \mapsto \frac{11}{2}$, $\bar{3} \mapsto \frac{14}{3}$, $6 \mapsto \frac{17}{3}$.

Theorem 6.2 (C -even leakage). *Per shared-link neighbor, in the C -even sector:*

$$\text{self-energy} = \text{hopping} = -\frac{481}{612} \quad \left(\text{channel split: } 1 : -\frac{1}{12}, 8 : -\frac{16}{51}, \bar{3} : -\frac{1}{6}, 6 : -\frac{2}{9} \right), \quad (18)$$

independent of the relative link orientation, with vacuum subtraction $+3/4$ per neighbor. The equality of hopping and self-energy is the operator identity $\langle G_{p'} | W_p = \langle G_p | W_{p'}$ (both equal $(\chi_p + \bar{\chi}_p)(\chi_{p'} + \bar{\chi}_{p'})\langle 0 | \sqrt{2}$). Assembling at $k=0$ in the uniform A_1^{++} projection with neighbor count $4(2d-3)$ in d spatial dimensions,

$$m_+(y) = 4\Delta^+(\frac{3y}{2}) + y^2 L(d) + O(y^3), \quad L(d) = 4(2d-3) \left(-\frac{503}{612} \right), \quad (19)$$

and in particular, for $d=3$,

$$\boxed{m_+(y) = \frac{8}{3} - y - \frac{9397}{1020} y^2 + O(y^3)} \quad \left(\frac{9397}{1020} = 9.2127 \dots \right). \quad (20)$$

Theorem 6.3 (C -odd leakage). *In the C -odd sector:*

- (i) Within-plaquette. $W(\chi - \bar{\chi})/\sqrt{2} = (\chi_{\bar{6}} - \chi_6)/\sqrt{2} + (\text{multiplet part})$; parity removes the vacuum and adjoint intermediates, leaving only the $20/3$ pair: the within-plaquette second order is $-1/4$, consistent with the restricted theorem via $9c_2^- = 9 \cdot \frac{1}{18} = \frac{1}{2}$.
- (ii) The self-energy is C -independent. The cross-component networks $\langle \chi_p | W_{p'} R W_{p'} | \bar{\chi}_p \rangle$ vanish identically by link balance, so the C -odd neighbor self-energy equals the C -even value $-481/612$.
- (iii) Signed hopping. The forced component structure splits the hopping into channel families, $N_{\text{mixed}} = -\frac{27}{68}$ (channels $1, 8$) and $N_{\text{like}} = -\frac{7}{18}$ (channels $\bar{3}, 6$), with $N_{\text{mixed}} + N_{\text{like}} = -\frac{481}{612}$ recovering the C -even value. The C -odd hopping is their difference and depends on the relative shared-link orientation $s = \pm 1$:

$$t^-(s) = \frac{5s}{612}, \quad (21)$$

and there is no vacuum-mediated path (parity).

Consequently the C -odd diagonal leakage at $d = 3$ is $12(-\frac{481}{612} + \frac{3}{4}) = -\frac{22}{51}$, giving the diagonal y^2 coefficient $\frac{1}{2} - \frac{22}{51} = \frac{7}{102}$, with the band (hopping) contribution bounded by $\pm 12 \cdot \frac{5}{612} = \pm \frac{5}{51}$. The full band minimization is a signed hopping problem on the plaquette adjacency graph (the uniform projection valid for C -even is not automatic) and is left open; the bound above already fixes the coefficient to $[-\frac{3}{102}, \frac{17}{102}]$.

Remark 6.4 (A nearly immobile excitation). $|t^-| = 5/612$ is smaller than $|t^+| = 481/612$ by a factor of 96, and the diagonal C -odd leakage nearly cancels against the vacuum subtraction. At this order the C -odd flux excitation is essentially immobile and almost undressed: its mixed- and like-channel amplitudes nearly cancel. This is, to our knowledge, a new structural observation about the strong-coupling scalar sector.

Assumption 6.5 (Linked-cluster step). The infinite-volume assembly in Theorem 6.2 excludes vacuum-disconnected second-order pieces by the standard linked-cluster theorem for quasi-degenerate strong-coupling expansions [2, 5]. The finite-system bookkeeping of these pieces (the vacuum path contributing $+3/4$ exactly once) is verified independently in Section 7; the infinite-volume cancellation itself is invoked, not re-proved here.

7 Independent verification: exact diagonalization on the domino

To test every constant of Section 6 without perturbative input, we exactly diagonalize (2) on the smallest lattice containing a neighbor pair: two plaquettes sharing one link (seven links, the “domino”). The gauge-invariant function space through the relevant degree is spanned by 19 loop and word functions (including the crossed words $\text{Tr}(U_\ell P U_\ell Q^\dagger)$ that complete the like-orientation channels); the Gram, magnetic, and electric matrices are computed in exact rational arithmetic by a Haar tensor-network contraction engine (Weingarten moments through $k = 2$, ε -tensor third moments), with the electric operator realized by Fierz insertions. All gates pass exactly: known Haar moments; exact self-adjointness of the electric matrix; Gram rank 19; the full $y = 0$ spectrum $\{0, \frac{8 \times 4}{3}, 4 \times 2, \frac{14 \times 2}{3}, \frac{17 \times 2}{3}, \frac{11 \times 2}{2}, 6 \times 2, \frac{20 \times 4}{3}\}$; and $E_{\text{vac}} \rightarrow -\frac{3}{2}y^2$.

Quasi-degenerate perturbation theory, using only the constants of Section 6 with honest finite-system vacuum-path bookkeeping, predicts the four level coefficients above the vacuum; the exact diagonalization, extracted by Richardson at $y = 10^{-3}, 2 \times 10^{-3}$ with 60-digit arithmetic, gives:

level	prediction	exact diagonalization	difference
C -even symmetric	$1769/3060 = 0.5781046$	0.5781130	8.4×10^{-6}
C -even antisymmetric	$13/20 = 0.6500000$	0.6500011	1.1×10^{-6}
C -odd symmetric	$31/68 = 0.4558824$	0.4558835	1.2×10^{-6}
C -odd antisymmetric	$17/36 = 0.4722222$	0.4722236	1.4×10^{-6}

All four agree at the Richardson truncation scale. The C -even antisymmetric level equals the pure within-plaquette value $13/20$ *exactly* — the structural signature that all neighbor leakage cancels there — and the two C -odd values were produced by the diagonalization *before* the C -odd assembly was performed, i.e. they were blind targets. This jointly verifies $-481/612$, the channel energies, the dimension lemma, the vacuum-path treatment, the C -independence of the self-energy, and the signed hopping $\pm 5/612$. The certificate is a single self-contained script with hard gates (Appendix A).

8 Numerical context at $\beta = 5.7$

Two independent numerical anchors place the analytic results in context; both are deliberately modest and fully caveated.

Leading-order Euclidean SCE ratio. With $u(\beta)$ the exact fundamental character coefficient of the one-link measure (Weyl-measure quadrature), the leading plaquette-tube expansion gives $m(0^{++})a = -4 \ln u$ and $\sigma a^2 = -\ln u$, hence $m/\sqrt{\sigma} = 4\sqrt{-\ln u}$, which evaluates to 3.82 at $\beta = 5.7$ and 3.71 at $\beta = 6.0$. The individual quantities are far outside their validity window at these couplings; only the ratio, in which the leading u -dependence cancels, is meaningful, and it coincides with the properly resummed strong-coupling value of Münster [5] (≈ 3.7). Its residual $\sim 10\%$ against the continuum 3.405(21) [8] is uncontrolled at this order.

Monte Carlo measurement. A small Wilson-action ensemble at $\beta = 5.7$ on $10^3 \times 16$ (8 chains $\times$ 540 measurements, 12 smeared A_1^{++} operators), analyzed with a $C(0)$ -metric fixed-vector generalized eigenvalue projection and tail subtraction (the pipeline validated on synthetic data with exact known answers before touching the ensemble), yields

$$a m(0^{++}) = 1.146 \pm 0.246 \quad (t=1; \text{upper estimate: excited-state contamination}), \quad (22)$$

consistent at 0.65σ with the finite- β literature value $aM(A_1^{++}) = 0.987(9)$ at $\beta = 5.6924$ [8], and $m/\sqrt{\sigma} = 2.82 \pm 0.61$ using the asymptotic $\sigma a^2 \simeq 0.165$ — i.e. squarely on the well-known Wilson-action scalar dip, below the continuum 3.405(21). Caveats: integrated autocorrelation times of 7–32 measurements make the quoted error optimistic by $O(10\text{--}30\%)$; single coarse coupling; no continuum statement. The three routes (Monte Carlo 2.8(6), finite- β lattice ≈ 2.4 , LO SCE ratio 3.7–3.8, continuum 3.405(21)) form a coherent picture: the dip is a lattice effect invisible to leading strong coupling.

9 Discussion

What has been established. A single compact operator — the one-plaquette class Hamiltonian — is now under exact analytic control across its entire coupling range, in both charge sectors, with closed-form general- N structure, a globally valid representation at the 10^{-3} level, an identification of its computational core with GUE Wick calculus, and a parameter-free falsifiable observable. By Theorem 5.1 this control transfers verbatim to the restricted 4D strong-coupling scalar channels, and the first correction beyond restriction is computed exactly in both sectors and verified by an independent exact diagonalization to one part in 10^6 .

Intended use. The most direct consumers are Hamiltonian formulations of lattice gauge theory: truncation schemes, tensor-network constructions, and quantum simulations of $SU(3)$ all rest on the plaquette spectrum computed here, and the exact towers, the global representation, and the verified leakage constants provide calibration targets at machine-checkable precision. The moment ratio $R_0(N)$ and the $SU(4)$ slope 14/17 are standalone falsifiable predictions requiring no commitment to the rest of the program.

What remains open — stated as open. (i) The infinite-volume linked-cluster step (Assumption 6.5). (ii) The C -odd signed-band minimization (bounded in Theorem 6.3). (iii) $O(y^3)$ leakage. (iv) A proven c_5 (numerical preview ≈ -0.0368). (v) A conformal-map uniformization of the crossover for a further order of magnitude in Theorem 4.2. None of these is required by any result above, and none is represented here as a step toward the continuum theory: as stated in Section 1.1, the methods of this paper do not reach the scaling window, and no claim about the Yang–Mills mass gap is made or implied.

A Verification inventory

Every quantitative claim in this paper is backed by at least two independent methods. The verification artifacts are single-file, self-contained scripts (exact rational arithmetic unless noted), each with hard gates that abort on failure:

- `ff_gap_rational.py` — rational second-quantized (Fock) Rayleigh–Schrödinger engine; reproduces c_0 – c_3 both sectors (validation gate) and certifies the general- N closed forms of Theorem 2.2 for $3 \leq N \leq 10$.
- `sun_gap_free_fermion_certificate_c3.py`, `su3_c4_exact_certificate.py` — exact $c_3(N)$ and $c_4^\pm$ theorems, with independent compact character-basis extractions agreeing to 1.05×10^{-4} and 9.1×10^{-6} relative.
- `gue_shell_calculus_verifier.py` — machine verification of Proposition 3.1 (α_N , $H_1 = -p_4/48$, γ_N).
- `r_moment_ratio_measurement.py` — Monte Carlo confirmation of $R_{\text{SU}(3)} = 1/24$ at the half-percent level.
- `domino_ed_certificate.py` — the exact diagonalization of Section 7: exact Haar tensor-network contraction engine (Weingarten $k \leq 2$, ε moments), Fierz-insertion electric operator, 19-function Gram of exact rank, full $y = 0$ spectral gate, and the four-coefficient comparison table, all as hard gates (~ 2 – 4 min CPU).
- `glueball_v4_reanalysis_fixedvec.py`, `glueball_v4_robustness.py` — the Monte Carlo pipeline of Section 8, with a mandatory synthetic self-test (exact known answer, matched storage precision) that must pass before real data is read.

Two methodological lemmas discovered through these gates are recorded for reuse: (i) with anti-Hermitian generators, the Casimir cross terms are $-\sum_a L_a \otimes L_a = +T \otimes T$ for like slot pairs but $-T \otimes T^T$ for mixed $(U, \bar{U})$ pairs — detectable via the identity $\int \Delta f = 0$; (ii) for noisy variational bases, whitening must use the positive-definite $C(0)$ metric: whitening at $t_0 \geq 1$ amplifies noise directions, and the maximum eigenvalue taken independently at each time separation rides the random-matrix noise edge and is biased low.

B Coefficient tables

Charge-even strong tower, orders 7–10 (exact).

$$\begin{aligned} b_7 &= -\frac{367979177879}{1344252672000000}, & b_8 &= +\frac{38764675528307}{8364238848000000}, \\ b_9 &= +\frac{1207377346074065219}{3794018741452800000000}, & b_{10} &= +\frac{3830311511546635473989}{92555087197741056000000000}. \end{aligned}$$

Two-sided control of $\Delta^+(\beta)$ (relative errors against the exact compact spectrum).

β	exact	strong tower	weak law
0.5	0.606072	-5×10^{-7}	(div.)
1.0	0.606004	-7×10^{-4}	-0.46
1.5	0.670085	-2.0×10^{-2}	-0.16
2.0	0.777023	-0.15	-4.3×10^{-2}
2.5	0.901247	(div.)	-6.5×10^{-3}
3.0	1.026448	(div.)	$+2.6 \times 10^{-3}$
8.0	1.958561	(div.)	$+4.7 \times 10^{-4}$

Leakage channel data (per shared-link neighbor, $E_0 = 8/3$).

channel	1	8	$\bar{3}$	6
norm $N_R = d_R/9$	1/9	8/9	1/3	2/3
energy	4	11/2	14/3	17/3
second-order term	-1/12	-16/51	-1/6	-2/9

Mixed-family sum $-27/68$; like-family sum $-7/18$; total $-481/612$.

LO Euclidean SCE with exact $u(\beta)$.

β	$u(\beta)$	ma (LO)	σa^2 (LO)	$m/\sqrt{\sigma}$ (LO)
4.0	0.27962	5.097	1.274	4.515
5.0	0.35395	4.154	1.039	4.076
5.7	0.40272	3.638	0.910	3.815
6.0	0.42253	3.446	0.862	3.713

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